

# Maintained reproduction of Coppock, Gross, Porter, Thorson and Wood (2023)

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*Drafted by Claude Opus 5 under the supervision of Alex Coppock.*

# 1 What this repository is

This repository re-runs the analysis behind a published article from its deposited data, in current R, and lays every number the article prints against what that analysis produces. It holds three things: a script that fetches and verifies the authors' replication archive, a rewrite of the analysis using current packages, and a ground truth that records, claim by claim, whether the published number reproduces.

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|                     |                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Article             | Coppock A, Gross K, Porter E, Thorson E and Wood TJ (2023). "Conceptual Replication of Four Key Findings about Factual Corrections and Misinformation during the 2020 US Election: Evidence from Panel-Survey Experiments." <i>British Journal of Political Science</i> .<br><a href="https://doi.org/10.1017/S0007123422000631">https://doi.org/10.1017/S0007123422000631</a> |
| Replication archive | <a href="https://doi.org/10.7910/DVN/WQXZUP">https://doi.org/10.7910/DVN/WQXZUP</a>                                                                                                                                                                                                                                                                                            |
| Pre-analysis plan   | <a href="https://osf.io/2hmv5">https://osf.io/2hmv5</a>                                                                                                                                                                                                                                                                                                                        |

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## 1.1 Folder layout

|                                    |                                                                             |
|------------------------------------|-----------------------------------------------------------------------------|
| <code>download_original.R</code>   | fetches and verifies the deposited archive                                  |
| <code>run_all.R</code>             | runs everything, in order                                                   |
| <code>original/</code>             | the deposited archive (not redistributed; fetched on demand)                |
| <code>original_manifest.csv</code> | every deposited file, its size and its published checksum                   |
| <code>maintained/</code>           | the rewrite; <code>maintained/output/</code> holds every number it produces |
| <code>ground_truth/</code>         | the extraction, the comparison and the gates                                |
| <code>errata.qmd</code>            | corrections to the published article                                        |

## 1.2 How to reproduce

Download the repository, open `coppock_etal_2023.Rproj`, and run:

```
source("run_all.R")
```

`run_all.R` fetches the deposited archive from Harvard Dataverse, verifies every file against its published checksum and byte size, fits every model, writes every figure and table, runs the deposited scripts in a scratch copy, rebuilds the ground truth and re-verifies the archive. Nothing writes into `original/`.

## 2 The paper

Eight panel-survey experiments fielded on Lucid and Mechanical Turk between September and November 2020, with 17,629 respondents across 24 experiments built on 21 distinct PolitiFact fact-checks. Each respondent saw three topics; for each topic they were assigned to pure control, misinformation, or misinformation followed by a fact-check. Belief certainty in the false claim is the primary outcome, measured again after at least a week.

The headline estimates are a meta-analytic misinformation effect of 4.3 points and a correction effect of -10.5 points on the 100-point belief certainty scale.

## 3 Does the deposited archive run?

The deposit is fourteen scripts, six data files and thirty-four saved model objects, in a directory tree of 63 files. Every script was run in a scratch copy, twice: once as the archive ships, and once stripped to data plus code so that a script which passes only because the deposit ships the object it reads is visible as a failure.

Warning in styling\_latex\_scale(out, table\_info, "down"): Longtable cannot be resized.

Table 1: Deposited scripts by pass and outcome.

| pass               | clean | error |
|--------------------|-------|-------|
| as_shipped         | 12    | 2     |
| stripped           | 7     | 7     |
| stripped_after_fit | 10    | 3     |

Two scripts fail in every pass, for reasons unrelated to the analysis. `appendix_figure_3.R` asks the graphics device for a font family that is not installed, and `appendix_figure_9.R` reads `data/clean/all_panels_long.rds`, a path the deposit does not contain: its copy of that file sits at `data/all_panels_long.rds`.

Stripping the copy to data plus code is what the second pass adds, and it changes the answer: 7 of 14 scripts fail without the deposited model objects, against 2 as shipped. Every one of the additional failures is a `readRDS` of an object `fit_all_models.R` builds. Running `fit_all_models.R` first and then the analysis scripts, which is the third pass in the table above, leaves 3 failures. Two are the pair that fail regardless. The third is `figure_5.R`, and it is a finding of its own: it reads `fitted_models/meta_trait_dic_w1_all_adj`, a deposited file with no extension, while the deposit's own code writes that object as `meta_trait_dic_w1_all_adj.rds`,

which the archive does not contain. The object survives only as a deposited artifact; nothing in the deposit can regenerate it under the name the figure reads.

**The deposit writes nothing except model objects.** Grepping the fourteen scripts for uncommented calls that touch a file returns thirty-four `write_rds()` lines, all in `fit_all_models.R`, and nothing else: no `ggsave`, no `write_csv`, no `sink`. Every figure is drawn to the active device and every table is printed to the console as LaTeX. So there is no deposited artifact to diff against, and a checksum result on `original/` is a statement about the download rather than about the analysis.

Running the deposit in place would not be harmless. Measured by modification time in a scratch copy, `fit_all_models.R` overwrites 32 deposited model objects and adds 3 more, and the figure scripts leave an `Rplots.pdf` behind.

### 3.1 Do the deposited objects follow from the deposited data?

Every model the deposit saves was refitted from the survey data and compared against the saved object.

Warning in `styling_latex_scale(out, table_info, "down")`: Longtable cannot be resized.

Table 2: Deposited model objects against a refit from the deposited data.

| objects | estimates agree to 1e-9 | estimates disagree | cells missing on one side |
|---------|-------------------------|--------------------|---------------------------|
| 34      | 23                      | 11                 | 200                       |

Warning in `styling_latex_scale(out, table_info, "down")`: Longtable cannot be resized.

Table 3: The objects where the refit and the deposit disagree, and what the deposit's own specification gives.

| object                                  | rows_deposited | rows_rewrite | estimate   | archive_specification |
|-----------------------------------------|----------------|--------------|------------|-----------------------|
| <code>ft_targets_ates_w1_PID_adj</code> | 138            | 132          | 28.7025088 | 0                     |
| <code>ates_w1_p3_adj</code>             | 6              | 6            | 12.8732396 | 0                     |
| <code>persistence_meta</code>           | 30             | 30           | 7.4147056  | NA                    |
| <code>ates_w3_p3_adj</code>             | 6              | 6            | 7.0610233  | 0                     |
| <code>ft_targets_ates_w1_all_adj</code> | 46             | 44           | 4.6340926  | 0                     |
| <code>ft_targets_meta_w1_PID_adj</code> | 6              | 6            | 2.9569308  | NA                    |
| <code>meta_trait_cates_w2_p2_adj</code> | 18             | 18           | 2.0977864  | NA                    |

|                             |    |    |           |    |
|-----------------------------|----|----|-----------|----|
| iv_p3_adj                   | 6  | 6  | 1.4617805 | 0  |
| ft_targets_meta_w1_all_adj  | 2  | 2  | 0.6069452 | NA |
| meta_trait_cates_w1_all_adj | 36 | 36 | 0.4848515 | NA |
| meta_trait_dic_w1_all_adj   | 18 | 18 | 0.2771084 | NA |

The `archive_specification` column is the same comparison run under the deposit's own model formulas rather than the corrected ones, and where it returns zero the disagreement above it is entirely the correction. The two thermometer objects are the exception and are discussed under the errata below.

Two properties of the comparison are worth stating because a single summary number hides both. The rewrite reverses the confidence interval the deposit prints backwards for every negated contrast, so a maximum taken over all six numeric columns at once is the width of the widest interval in the object and says nothing about agreement. And a cell one side computes while the other returns as missing is invisible to any difference, which is exactly how the political knowledge estimates move: the two `estimatr` versions disagree about which HC2 standard errors are estimable, so the pooling drops a different study.

## 4 Errata

Six sentences in the published article are wrong, and they are corrected in [coppock\\_etal\\_2023\\_errata.pdf](#), whose values are computed at render time from this repository's output. None of them changes a conclusion. In summary: the reported total sample size is 17,681 where the article's own appendix table sums to 17,629; the count of experiments in which misinformation significantly reduced accuracy is given as twelve where the deposited estimates give 13; two different appendix floats are both captioned Table 1; and the four sentences reporting how much of the correction effect survives to a later wave report a figure of the order a fixed-effect pool gives where the appendix gives a random-effects pool of the same ratios, 48.9 per cent at Wave 2 and 34.3 per cent at Wave 3.

### 4.1 Corrections the rewrite makes to the deposited code

Four coding errors in the deposit are corrected in `maintained/`. Each is unambiguous, in the sense that the code plainly does something other than what the surrounding code says it is doing.

**The thermometer models are fitted on the wrong sample.** The deposit's loop over targets reads

```
filter(all_panels_long, fc == fc, topic_short_2 == topic_short_2)
```

Inside `filter()` the bare names resolve to the columns, so both conditions compare a column with itself and neither subsets anything: every target model is fitted on all eight panels pooled, including respondents who were never shown the claim. The loop extracts `fc` and `topic_short_2` one row at a time in order to subset, so the intent is not in doubt. Applying the subset costs one of the twenty-three target rows, because Panel 3 names Barack Obama as a target and never asked the Barack Obama thermometer; that row is recorded in `maintained/output/fits/ft_targets_unfittable.csv`.

This moves Figure 6. The published overall labels are -0.19 and 0.18; the corrected estimates are -0.74 and 0.8. All eight of the figure’s labels change, and with them two sentences of prose: the introduction’s “smaller than one-quarter of a point” and the Results section’s “smaller than half a point” are both true of the published figure and false of the corrected one.

There is a second thing wrong with Figure 6 and it is not the rewrite’s doing. Six of its eight labels are exactly what the deposited objects give. The two Overall labels are not: the deposit returns -0.23 and 0.19 where the page prints -0.19 and 0.18. The figure script reads those two labels straight out of `ft_targets_meta_w1_all_adj.rds` and rounds them to two decimals, so the published pair cannot be recovered from the archive as deposited. The cause is not established.

These stay findings here rather than errata. The published cells reproduce from the deposit exactly, so what is wrong with them is a matter of method rather than of arithmetic, and correcting a column of cells the deposit reproduces exactly is a different act from correcting a misstated number. The substantive claim is untouched either way: a correction effect of 0.8 points on a 100-point feeling thermometer is as close to nothing as 0.18 is.

**The adjusted three-wave models regress the wrong outcomes.** In the deposit, `ates_w1_p3_adj` and `ates_w2_p3_adj` both regress `outcome_w3`, `ates_w3_p3_adj` regresses `outcome_w2`, and `iv_p3_adj` instruments `outcome_w2`. The unadjusted block immediately above pairs each wave with its own outcome, and so does the covariate-adjusted conditional-effect block below, so this is a copy-and-paste slip. Two published artifacts show it. Appendix Table 7’s `w1` row holds the wave 3 estimate and its unlabelled row holds the wave 2 estimate, which is why 7 of its 21 cells reproduce where the other six regression tables reproduce completely. And Figure 4’s bottom-left panel draws its Wave 1 and Wave 2 points at the same height, because they are the same estimate.

**Every negated confidence interval prints backwards.** The deposit’s `filter_and_flip()` negates the estimate and both interval bounds for the misinformation contrast without reversing their order, so those cells print as `[upper, lower]`. 144 of the 492 published appendix cells are printed this way. The interval itself is right; only the order is wrong. The rewrite swaps the bounds back.

**The significance column of the appendix regression tables is not a comparison of numbers.** The deposit's table script reshapes each fit with `reshape2::melt`, and where a fit carries a character column outside `id.vars` every melted value becomes a string. The surviving `if_else(p.value < 0.05, "* ", " ")` then compares two strings, so "0.0144" sorts before "0.05" and earns a star while "1.28e-10" sorts after it and does not. The effect is that the most significant estimates in the table are the ones printed without a marker: the published tables carry 205 stars where 256 cells have  $p < 0.05$ . This is deterministic, it is visible on the page once you know to look for it, and it is a property of the presentation rather than of any estimate, so it stays a finding here rather than an erratum.

## 4.2 The persistence percentages are pooled one way in the main text and another in the appendix

The article states three times that fact-check effects persist “at 66 per cent” of their original magnitude after one week, once at the sharper “66.4 per cent”, and “at 50 per cent” after more than two weeks. The appendix reports the same quantity and gives different numbers.

The quantity the pre-analysis plan defines, and the one the appendix's own persistence tables print in their `ratio` rows, is the per-experiment ratio of the later-wave coefficient to the Wave 1 coefficient, pooled across experiments by meta-analysis. Pooled with random effects, which is what `metafor` does by default and what the appendix's meta-analysis rows carry, it is 48.9 per cent at Wave 2 and 34.3 per cent at Wave 3 with covariates, and 54.4 and 38.9 per cent without them. Pooled with fixed effects, the same ratios give 64.9 and 70.8 per cent at Wave 2 and 52.2 and 63.3 per cent at Wave 3. The ratio of the pooled Wave 2 effect to the pooled Wave 1 effect, a different derivation of the same idea, gives 46.9 per cent, and the corresponding Wave 3 figure is 39.7 per cent.

The published percentages are of the order the fixed-effect pool gives and not of the order the appendix's own rows give, so a reader of the introduction takes away a materially different number from a reader of the appendix. No combination of adjustment, estimator and sample returns 66.4 or 50 exactly. The article should have used random effects throughout, and the four sentences carrying these figures are corrected on that basis in `errata.qmd`. The Wave 3 correction is larger than the pooling alone would make it, because the ratio behind appendix Table 7 also instruments the wrong outcome; see the three-wave finding above.

## 4.3 Two further quantities with no counterpart

The appendix states that attitudinal outcomes were observed for 22 political figures and groups. The deposited data carry 14 feeling thermometer columns. And appendix Table 9 reports a mean adjusted r-squared for each of ten covariates across “our 21 experiments, resulting in 210 separate linear models”; the deposit ships no code for that table, and the

article elsewhere describes 24 experiments. The rewrite adds no script for it, since writing one would be estimating something the deposited archive never did.

## 5 The ground truth

`ground_truth/coppock_etal_2023_ground_truth.csv` carries one row per published claim. `value_paper` is the string the article prints and is the only column anyone typed; `value_script` is read out of the deposit and `value_rewrite` out of `maintained/output/`, both by script. A value agrees when the computed number, printed to the page's own precision, gives the same digits.

Warning in `styling_latex_scale(out, table_info, "down")`: Longtable cannot be resized.

Table 4: Published claims by type and verdict.

| claim_type   | does not reproduce | no verdict | reproduces | does not hold | holds |
|--------------|--------------------|------------|------------|---------------|-------|
| definitional | 3                  | 9          | 21         | 0             | 0     |
| descriptive  | 0                  | 2          | 0          | 3             | 15    |
| pipeline     | 21                 | 11         | 130        | 0             | 0     |
| structural   | 1                  | 12         | 12         | 0             | 0     |
| transcribed  | 0                  | 14         | 0          | 0             | 0     |

Warning in `styling_latex_scale(out, table_info, "down")`: Longtable cannot be resized.

Table 5: Where the fault lies, on every row that is not a clean match.

| defect_locus   | rows |
|----------------|------|
| archive        | 26   |
| environment    | 3    |
| paper_internal | 8    |
| unresolved     | 2    |

Every row that is not a clean match carries a `defect_locus`, and no clean match carries one. The five values are `paper_internal` (the article disagrees with its own tables or its own data), `archive` (the deposit cannot support the claim), `environment` (R or a package moved underneath it), `rewrite` (ours) and `unresolved` (cause not established).

## 5.1 Coverage, float by float

Warning in styling\_latex\_scale(out, table\_info, "down"): Longtable cannot be resized.

Table 6: Coverage per published float. Printed numbers are what a float puts on its own face; plotted quantities are what it draws.

| float                                 | printed numbers | covered | fraction | plotted quantities | counts v |
|---------------------------------------|-----------------|---------|----------|--------------------|----------|
| Appendix Figure 1                     | 0               | 0       | NA       | 36                 |          |
| Appendix Figure 2                     | 0               | 0       | NA       | 72                 |          |
| Appendix Figure 3                     | 0               | 0       | NA       | 21                 |          |
| Appendix Figure 7                     | 0               | 0       | NA       | 48                 |          |
| Appendix Figure 8                     | 0               | 0       | NA       | 108                |          |
| Appendix Figure 9                     | 12              | 1       | 0.083    | 12                 |          |
| Appendix Figures 4 to 6               | 0               | 0       | NA       | 816                |          |
| Appendix regression table 1           | 400             | 400     | 1.000    | 400                |          |
| Appendix regression table 2           | 400             | 400     | 1.000    | 400                |          |
| Appendix regression table 3           | 400             | 400     | 1.000    | 400                |          |
| Appendix regression table 4           | 300             | 300     | 1.000    | 300                |          |
| Appendix regression table 5           | 300             | 300     | 1.000    | 300                |          |
| Appendix regression table 6           | 84              | 84      | 1.000    | 84                 |          |
| Appendix regression table 7           | 84              | 28      | 0.333    | 84                 |          |
| Appendix Table 1 (sample composition) | 206             | 9       | 0.044    | 206                |          |
| Appendix Table 8                      | 72              | 0       | 0.000    | 72                 |          |
| Appendix Table 9                      | 10              | 1       | 0.100    | 10                 |          |
| Figure 1                              | 0               | 0       | NA       | 72                 |          |
| Figure 2                              | 72              | 72      | 1.000    | 72                 |          |
| Figure 3                              | 68              | 0       | 0.000    | 104                |          |
| Figure 4                              | 0               | 1       | NA       | 30                 |          |
| Figure 5                              | 36              | 36      | 1.000    | 36                 |          |
| Figure 6                              | 8               | 8       | 1.000    | 8                  |          |

Seven of the appendix's floats and two of the article's print no numbers at all. For those the checkable published quantity is the count of what they plot, and that count is verified; the fraction is left empty rather than reported as zero.

Two floats are covered thinly and both are worth stating plainly. **Figure 3 is the largest gap in this repository.** It labels every effect significantly different from zero and every one of its eight meta-analytic summaries, roughly 68 printed numbers, and none of those labels was transcribed. What is verified is that the figure plots 96 conditional effect estimates, and

that those estimates are the covariate-adjusted cells of appendix Tables 2 and 3, every one of which reproduces. **Appendix Table 1** prints 206 numbers, of which the eight panel sizes are checked individually and the rest are checked as counts of non-empty cells.

The seven appendix regression tables are the bulk of the published record: 492 cells, each printing an estimate, a standard error and two interval bounds, so 1968 published numbers. All of them are parsed positionally from the published PDF into `ground_truth/published_appendix_values.csv` and compared cell by cell.

## 6 The extraction and the two instruments

`ground_truth/published_claims.csv` is the extraction: 254 rows, one per numeric claim in the article and its appendix, each classified by hand as `pipeline`, `descriptive`, `definitional`, `structural` or `transcribed`, and each carrying the string the page prints and the precision it prints it at. Spelled-out numbers were swept for separately from digits, since no token scan sees “eight panel experiments”, “twelve out of the twenty-four opportunities” or “thirteen days”; 20 of the extraction’s quoted sentences state their number in words.

Two instruments read the same pipeline output by separate paths. `ground_truth/build_ground_truth.R` builds the comparison table. `maintained/in_text_claims.R` carries the article’s own sentence in a block comment beside code that recomputes the number, and prints 219 claims in the form `CLAIM <id> = <value> || <label>`. It reads `maintained/output/` and the extraction, never the ground truth, so the two derivations are independent and a disagreement between them is a finding.

The build asserts all of the following and stops if any fails.

- Every `pipeline` and `descriptive` claim has a block and a row.
- The number of printed claims equals the number of rows requiring one, and the two sets of ids are equal in both directions.
- Every value the two instruments both produce agrees exactly.
- Re-rendering `value_paper` from its own number at its own recorded precision returns the string the extraction stores, which is what catches a precision that is right about the value and wrong about the digits.
- Every adverse row carries a `defect_locus` and no clean match does.
- Every published float has at least one row.

Each of the first, third and fourth was tested by breaking it: deleting a block fails the count, perturbing a printed value fails the cross-check, and changing one claim’s recorded precision fails the string check.

## 7 The maintained rewrite

`maintained/` is 22 scripts. `fit_all_models.R` fits every model from the deposited survey data and writes each family as a tidy CSV under `maintained/output/fits/`, so every number a figure or table prints can be traced to a file a reader can open. Nothing downstream reads the deposit's saved objects.

The substitutions the rewrite makes are the usual ones for code of this age: the native pipe for `magrittr`, `pivot_longer/pivot_wider` for `reshape2::melt/dcast`, `reframe(tidy(...))` for `do(tidy(...))`, `pick(everything())` for `cur_data()`, `write_csv()` for `xtable` printed to the console, `linewidth` for the deprecated `size` aesthetic, and `position_dodge()` for `ggstance::position_dodgev()`. The Roboto font dependency in the traffic figure is dropped. `plyr` is not loaded at all, which removes the masking hazard the deposit works around by loading it first.

## 8 Figure verification

Every rendered figure was laid beside the published one. Numbers alone cannot catch a transposed axis or a mislabelled series, and three of this article's six figures print their estimates on the face of the plot, which makes them published tables in disguise.

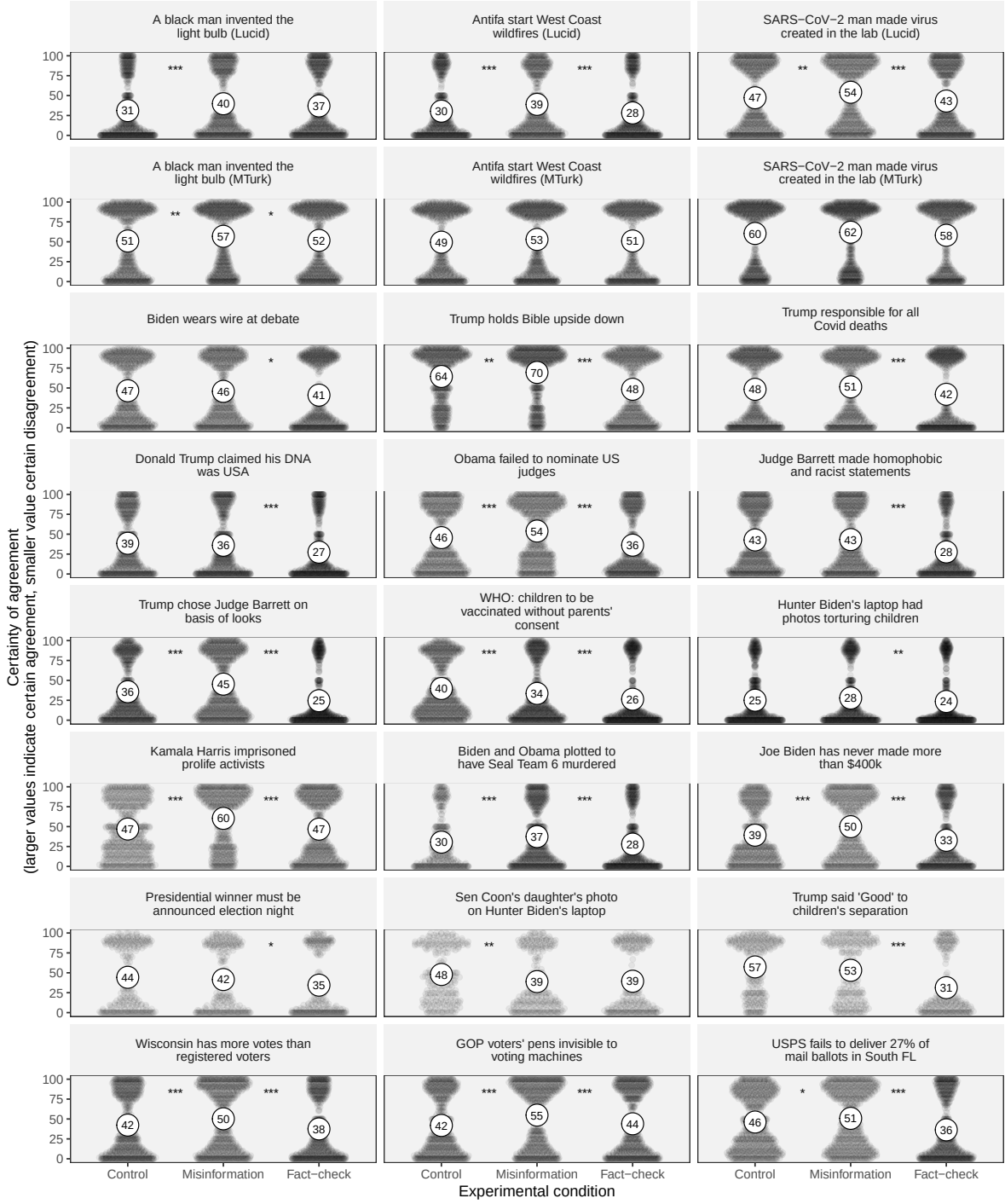


Figure 1: Figure 2 as the rewrite draws it. All seventy-two conditional means reproduce the published labels exactly.

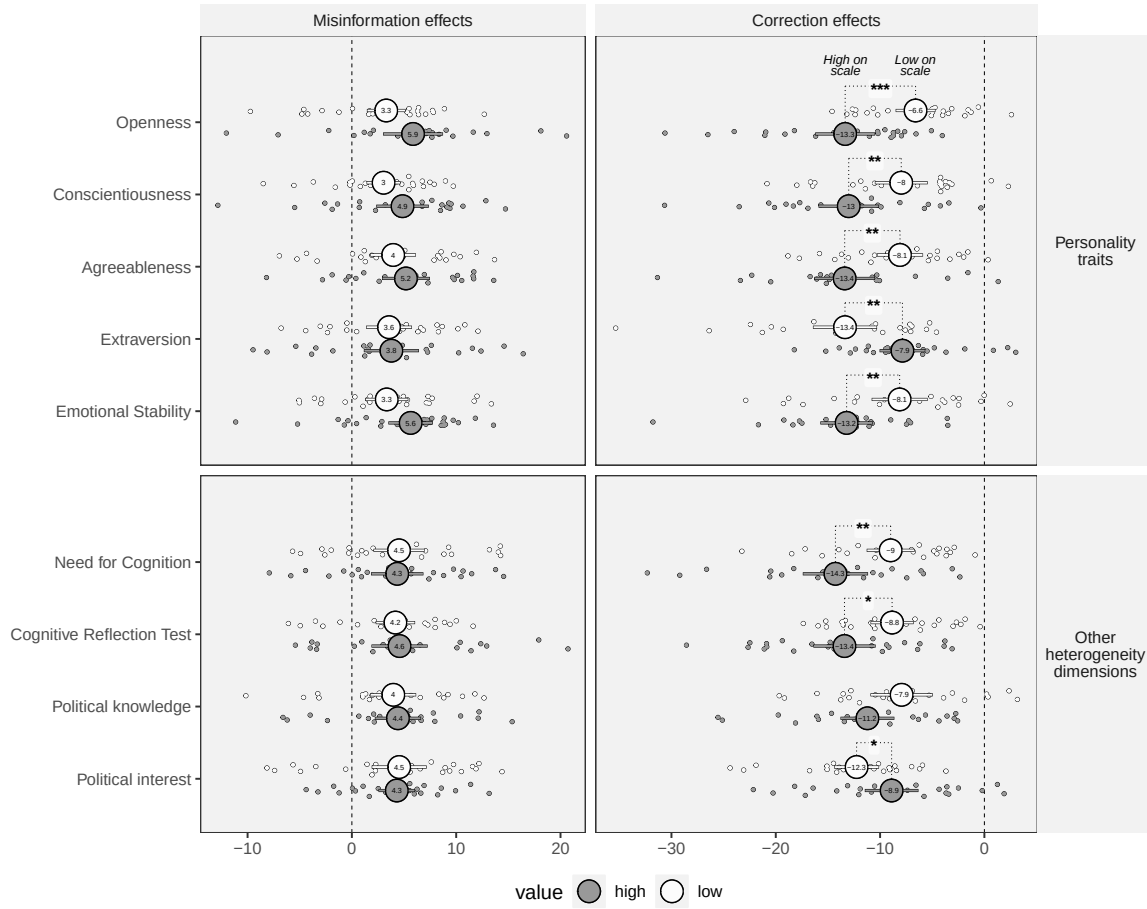


Figure 2: Figure 5 as the rewrite draws it. Thirty-four of thirty-six labels reproduce; the two political knowledge correction labels move, which is environment drift rather than a defect in either analysis.

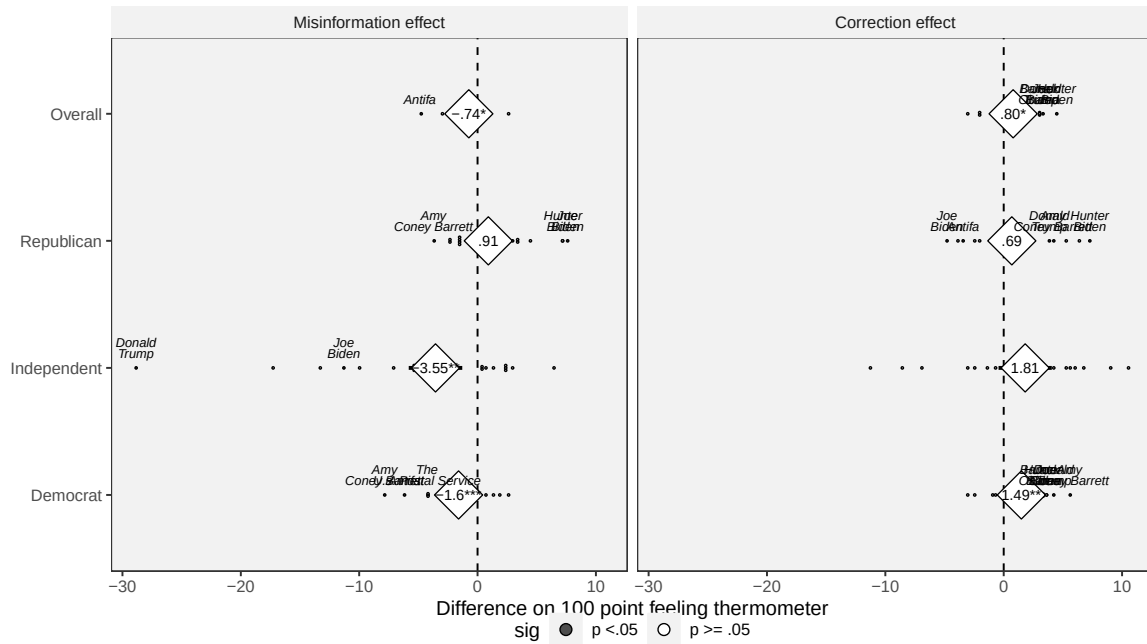


Figure 3: Figure 6 as the rewrite draws it, with the thermometer models fitted on the panel that carried each claim. All eight labels differ from the published ones.

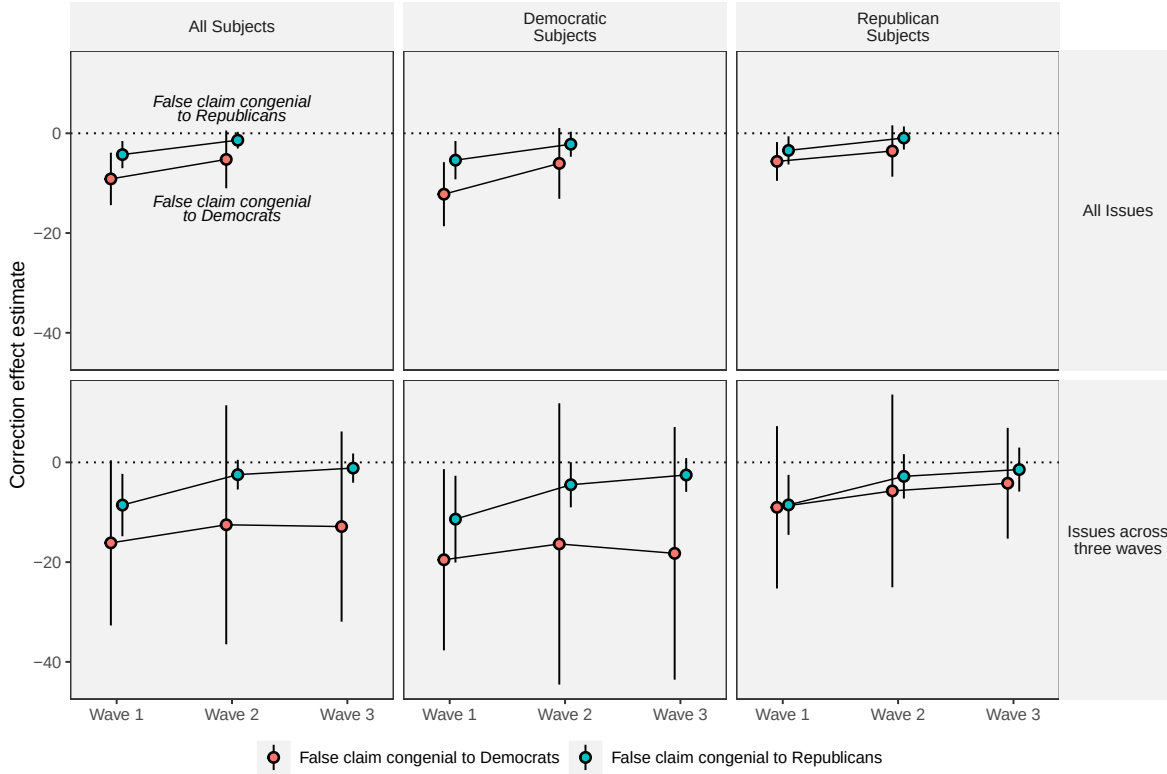


Figure 4: Figure 4 as the rewrite draws it. In the published version the Wave 1 and Wave 2 points of the bottom-left panel sit at the same height, because the deposit fits both on the Wave 3 outcome.

## 9 Rewrite verification

`run_all.R` was run to completion twice from clean sessions and the whole tree diffed. Every CSV is byte-identical between runs. The figure PDFs differ, because a PDF records the time it was written.

`original/` is verified twice per run: `download_original.R` is sourced first, which checks every file against the MD5 Dataverse serves for `?format=original` and against its deposited byte size, and refuses to continue if `original/` holds any file the manifest does not list, dotfiles included. It is sourced again as the last step, because the first pass proves only that the archive was intact when the run began. All 63 files match, and 0 carry a published checksum that disagrees with the bytes Dataverse serves. Checksums for files this repository writes are not quoted anywhere: a PDF records its write time, so its hash changes on every run.

## 10 R environment

Warning in styling\_latex\_scale(out, table\_info, "down"): Longtable cannot be resized.

| component | version |
|-----------|---------|
| R         | 4.6.0   |
| tidyverse | 2.0.0   |
| estimatr  | 1.0.6   |
| metafor   | 5.0.1   |
| ggplot2   | 4.0.3   |
| broom     | 1.0.13  |

The rewrite keeps the current sampler and the current packages throughout. Nothing in this analysis draws at random: no script calls `sample()`, `rnorm()` or a bootstrap, so the results are bit-identical at any seed and the sampler change in R 3.6 is not in play.